

BUFFALO HARBOR, NY, USA

WMO: 997254

Lat: 42.878N

Lon: 78.890W

Elev: 178

StdP: 99.20

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.4	-13.8	N/A	N/A	N/A	N/A	N/A	N/A	18.0	-1.9	15.9	-2.4	5.9	270	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	26.8	N/A	25.8	N/A	25.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.6	12.3	10.8	DB	-18.6	29.3	3.9	1.3	-21.4	30.2	-23.6	31.0	-25.8	31.7	-28.6	32.6	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-3.7	-4.1	-0.1	6.5	12.7	18.6	21.7	21.2	17.6	11.8	5.6	0.4	(6)
(7)		DBStd	10.05	5.93	5.42	4.96	4.04	4.03	3.01	2.29	2.23	3.29	4.02	4.58	4.58	(7)
(8)		HDD10.0	1743	426	396	314	119	18	0	0	0	0	27	145	298	(8)
(9)		HDD18.3	3653	683	629	571	356	181	32	3	3	52	206	383	556	(9)
(10)		CDD10.0	1409	0	0	2	14	101	258	363	348	228	83	12	1	(10)
(11)		CDD18.3	279	0	0	0	0	6	40	107	93	30	3	0	0	(11)
(12)		CDH23.3	793	0	0	0	2	11	86	338	272	81	3	0	0	(12)
(13)		CDH26.7	39	0	0	0	0	1	9	14	12	4	0	0	0	(13)
(14)	Wind	WSAvg	4.7	6.1	5.4	4.5	4.1	3.5	3.6	3.9	4.0	4.1	5.1	5.6	6.1	(14)
(15)	Precipitation	PrecAvg	1029	73	76	79	78	80	95	68	93	101	85	109	98	(15)
(16)		PrecMax	1248	142	154	152	128	175	161	139	168	196	143	243	160	(16)
(17)		PrecMin	821	25	9	20	41	10	29	22	13	35	21	35	48	(17)
(18)		PrecStd	113	36	42	36	25	38	36	35	40	43	38	50	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	9.6	16.2	20.8	25.0	27.8	28.4	28.0	27.0	23.4	17.4	13.0	(19)
(20)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	8.7	6.6	12.3	17.4	22.2	25.5	26.7	26.5	25.3	21.2	15.4	10.2	(21)
(22)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	6.5	4.6	9.1	15.1	20.5	24.1	25.8	25.6	24.2	19.7	13.8	8.6	(23)
(24)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	4.3	2.6	6.6	12.9	18.9	22.9	25.1	24.9	23.0	17.9	12.4	6.7	(25)
(26)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)
(28)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range	MDBR	6.1	6.2	6.4	7.3	6.6	5.4	4.8	5.3	6.5	6.0	5.7	5.1	(35)	
(36)		5% DB	MCDBR	8.0	7.9	9.7	11.6	9.2	8.1	5.5	6.1	7.1	8.2	7.4	6.8	(36)
(37)		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.328	0.334	0.375	0.404	0.437	0.437	0.427	0.380	0.362	0.315	0.299	(40)	
(41)		taud	2.395	2.301	2.396	2.325	2.285	2.227	2.240	2.258	2.370	2.409	2.507	2.459	(41)	
(42)		Ebn at Noon	838	863	909	891	870	840	834	831	848	818	819	807	(42)	
(43)		Edh at Noon	78	101	105	121	130	138	135	129	107	91	70	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.30	2.10	3.26	4.40	5.39	5.81	5.86	5.21	4.14	2.51	1.55	1.04	(44)	
(45)		RadStd	0.12	0.21	0.36	0.47	0.56	0.44	0.46	0.29	0.38	0.24	0.22	0.14	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page